

Requirement Gathering Document – Talvix

Date: 22 July 2026

Project Overview

Project Title: Talvix – AI-Powered Skills-Based Recruitment and Assessment Platform

Problem Statement:

Traditional recruitment relies heavily on resumes and manual screening, making it difficult to identify candidates based on actual skills. Students struggle to find suitable opportunities and recruiters spend significant time reviewing applications and conducting assessments.

Objectives:

- Build an AI-powered recruitment platform.
- Enable skills-based hiring.
- Automate candidate screening and assessment.
- Provide coding tests, MCQs, hackathons, and technical evaluations.
- Match candidates to jobs using AI.
- Offer career recommendations.
- Detect fraudulent job postings.

System Scope

Talvix is proposed as a full-scale implementation developed as an MCA major project. The system includes Candidate, Recruiter, Organization, and Admin portals, along with an Assessment Engine, Communication Module, Analytics Dashboard, and AI-powered features such as Resume Skill Extraction, Candidate–Job Matching, AI Question Generation, Performance Analysis, Career Recommendation, and Job Scam Detection.

Target Audience

- Students / Job Seekers
- Recruiters
- Organizations / Companies
- System Administrators
- Educational Institutions (future use)

Modules

1. Authentication & User Management
2. Candidate Management
3. Recruiter & Organization Management
4. Job Management
5. Application Management
6. Assessment Engine
7. AI Intelligence Module

- 8. Communication Module
- 9. Analytics Dashboard
- 10. Admin Module

User Roles

Candidate – Apply for jobs, attend assessments, manage profile.
Recruiter – Post jobs, manage candidates and assessments.
Organization Admin – Manage recruiters and organization.
System Administrator – Full platform administration.

System Ownership

Academic Major Project
Department of Computer Applications
Amal Jyothi College of Engineering
Developed by: Geo George (Integrated MCA Student)

Industry / Domain

Industry: Human Resource Management (HRTech)
Domain: Artificial Intelligence, Recruitment Technology, Career Development, EdTech

Data Collection Contacts

1. Allen Shaji
CEO Leopard tech Lab Pvt
Leopard Tech Lab Pvt
Phone Number: 8547933472
LinkedIn: <https://www.linkedin.com/in/allen-shaji/>
Purpose: Recruitment workflow and hiring challenges.

2. Dr. Tom Sunny
Ph.D., Chief Innovation Officer, Centre for Research, Innovation & Consultancy (CRIC), and
Placement Officer at Amal Jyothi College of Engineering (Autonomous).
Email: placement@amaljyothi.ac.in,
Phone: +91 88482 48589,
LinkedIn: <https://www.linkedin.com/in/tom-sunny-phd/>,
AJCE Placement Cell: <https://www.ajce.in/home/placementcell.html>
Purpose: Campus recruitment process.

3. Solaman M J
Student at Amal Jyothi Engineering College
LinkedIn : <https://www.linkedin.com/in/solaman-m-j-6ba47331b/>
Phone Number: 8281530386
Purpose: Job search challenges and platform expectations.

Questionnaire for Data Collection

Candidate Questionnaire with Responses

Q1. How do you currently search for jobs?

The respondent stated that jobs are primarily searched through LinkedIn, campus placement drives, company career portals, and popular job portals.

Q2. What challenges do you face while applying for jobs?

The respondent mentioned that identifying suitable job opportunities, completing repetitive application processes, and not receiving feedback after applications are the major challenges.

Q3. Have you ever been rejected without receiving feedback?

The respondent confirmed that rejection without any explanation is a common experience during the recruitment process.

Q4. Which factors do you consider most important during recruitment?

According to the respondent, practical skills, project experience, technical knowledge, and communication skills are the most important factors considered by recruiters.

Q5. Have you attended online coding assessments?

The respondent stated that they have participated in online coding assessments conducted through platforms such as HackerRank and HackerEarth.

Q6. Would AI-based career recommendations be helpful?

The respondent agreed that personalized AI-based career recommendations would help identify suitable job opportunities and improve career planning.

Q7. Would a skill-gap dashboard be useful?

The respondent believed that a skill-gap dashboard would be beneficial for identifying weaknesses and planning future learning activities.

Q8. Have you encountered fake job postings?

The respondent indicated that fake or misleading job postings are frequently found on unverified online platforms and social media.

Q9. Which features would you like in a recruitment platform?

The respondent suggested features such as AI-powered job matching, coding assessments, verified company profiles, interview scheduling, and career guidance.

Q10. Would you trust AI-based job recommendations?

The respondent stated that AI-generated recommendations would be trustworthy if they are transparent, accurate, and based on verified candidate skills and qualifications.

Recruiter Questionnaire with Responses

Q1. Can you describe your recruitment process?

The recruiter explained that the recruitment process typically consists of job posting, resume screening, candidate shortlisting, technical assessments, interviews, and final selection.

Q2. What is the biggest challenge during resume screening?

The recruiter identified handling a large volume of applications while accurately identifying skilled candidates as the primary challenge.

Q3. Approximately how many resumes do you receive for a single job posting?

The recruiter stated that the number of applications generally ranges from 100 to 500 depending on the position and organization.

Q4. How much time is required for shortlisting candidates?

The recruiter mentioned that the shortlisting process may take several hours to multiple days depending on the number of applications received.

Q5. Do you use online assessments during recruitment?

The recruiter confirmed that online technical and coding assessments are commonly used to evaluate candidates before interviews.

Q6. Would AI-assisted resume screening be beneficial?

The recruiter agreed that AI-assisted screening would significantly reduce manual effort and improve recruitment efficiency.

Q7. What candidate information is most valuable during recruitment?

According to the recruiter, technical skills, project experience, assessment scores, and communication abilities are the most valuable evaluation criteria.

Q8. Would AI-powered candidate-job matching improve recruitment?

The recruiter stated that AI-based matching would improve hiring accuracy by recommending candidates whose skills closely align with job requirements.

Q9. What analytics would you like to see in a recruitment platform?

The recruiter expressed interest in analytics related to application trends, hiring funnel performance, assessment statistics, and candidate rankings.

Q10. What additional features would improve the recruitment process?

The recruiter recommended features such as interview scheduling, automated candidate communication, AI-based recommendations, and job fraud detection to enhance the overall recruitment experience.

Expected Outcome

The gathered requirements will identify user needs, existing recruitment challenges, functional and non-functional requirements, AI feature expectations, and future enhancements, forming the foundation for Talvix development.

